

Mohith Raagesh B

mohithbalakumar12@gmail.com | +91 94876 04349 | [LinkedIn](#) | [GitHub](#)

PROFESSIONAL SUMMARY

Full-Stack Developer with 1 year of professional experience and 3+ years of freelance development carried out during my studies, building scalable web and mobile applications using Node.js, Express.js, Next.js, React.js, Flutter, and MongoDB. Completed B.Tech in Computer Science and Engineering from Vellore Institute of Technology, Chennai in May 2025. Experienced in designing and shipping production-grade REST APIs, real-time systems using Socket.IO, server-side rendering with Next.js, and cross-platform mobile development with Flutter. Known for end-to-end product ownership, clean architecture, and the ability to independently deliver complete, production-ready full-stack applications.

EDUCATION

B.Tech in Computer Science and Engineering

May 2021 - May 2025

Vellore Institute of Technology, Chennai, India | GPA: 7.44 / 10

CBSE Class XII

May 2019 - May 2021

National Model Senior Secondary School, Coimbatore, India | 80%

TECHNICAL SKILLS

Languages: JavaScript (ES6+), Python, SQL

Backend: Node.js, Express.js, REST API Design, JWT Authentication, Socket.IO, MVC Architecture

Frontend: Next.js, React.js, HTML5, CSS3

Mobile: Flutter (Cross-platform, Android and iOS)

Database: MongoDB, Mongoose ODM, Schema Design, Indexing, PostgreSQL (Basics)

Caching: Redis

Real-time: Socket.IO, WebSockets

Tools: Git, GitHub, Postman, MongoDB Compass, VS Code

Concepts: Full Stack Development, REST API Design, System Architecture, Server-Side Rendering (SSR), Real-time Systems, Scalability, API Testing, Error Handling, Deployment

EXPERIENCE

Backend Developer, MotoDoctor Private Limited

Jul 2025 - Present

Chennai, India

- Designed and developed 15+ RESTful API endpoints using Node.js and Express.js, powering core application features used daily in production.
- Architected MongoDB collections and Mongoose schemas with robust validation, indexing, and relational data modeling to support scalable, high-volume data access.
- Implemented centralized error handling and input validation middleware, significantly improving API reliability and reducing integration friction for frontend teams.
- Collaborated with frontend developers to deliver end-to-end product features, using well-documented API contracts and Postman test suites to streamline delivery.
- Maintained high code quality through a structured MVC architecture, ensuring clear separation of concerns and long-term codebase maintainability.

Freelance Full-Stack Developer, Self-Employed

2022 - 2025

Remote (while studying at VIT Chennai)

- Independently designed, developed, and deployed multiple production-ready full-stack web and mobile applications over 3 years while completing my B.Tech degree.
- Architected complete systems end to end, from database schema design and REST API development to frontend UI and production deployment, demonstrating strong ownership across the full stack.
- Built cross-platform mobile applications using Flutter, delivering smooth and consistent user experiences across Android and iOS from a single codebase.

- Engineered real-time bidirectional communication and live tracking systems using Socket.IO and WebSockets, enabling sub-second data updates between clients and servers.
- Implemented automated billing and GST-compliant invoice generation, eliminating manual workflows and reducing provider effort to zero.
- Designed and integrated scalable REST APIs using Node.js and Express.js with optimized Mongoose schemas to handle concurrent read and write operations reliably.

Software Development Intern, Mandy Technologies Private Limited

Oct 2023 - Dec 2023

Sitra, Coimbatore

- Built responsive, cross-browser-compatible UI components using HTML, CSS, and JavaScript.
- Assisted in backend integration with MongoDB, gaining practical experience in full-stack data flow.
- Performed functional testing and debugging to ensure application reliability before release.

PROJECTS

Pawzman - Pet Services Mobile App | *Flutter, Node.js, Express.js, MongoDB, Socket.IO, JWT*

- Independently architected and deployed a production-grade full-stack mobile application, owning the complete stack from Flutter UI through to Node.js backend, MongoDB database, and live deployment.
- Implemented real-time bidirectional chat and live GPS-based booking tracking using Socket.IO and WebSockets, enabling seamless and reliable communication between users and service providers.
- Built a dynamic billing engine with automated invoice generation and GST calculation, serving providers with zero manual intervention and full audit trail.
- Designed MongoDB schemas using Mongoose with optimized indexing for efficient querying across user, provider, booking, and invoice collections at scale.
- Enforced JWT-based authentication, structured input validation, and centralized error handling to deliver a secure, reliable, and maintainable API layer.

ESpot - EV Route Planning Web App | *Next.js, React.js, Node.js, Express.js, MongoDB, Mapbox API*

- Built a full-stack web application using Next.js and Node.js enabling EV drivers to plan optimized routes with real-time visualization of charging station infrastructure.
- Leveraged Next.js server-side rendering (SSR) for fast initial page loads, improved SEO, and better performance on low-bandwidth connections.
- Developed backend route-calculation logic factoring in vehicle range, travel time, distance, and charging stop availability, demonstrating practical system architecture thinking.
- Integrated Mapbox APIs for geolocation, turn-by-turn routing, and distance matrix calculations across dynamic route data.
- Implemented fallback and retry mechanisms for external API failures, ensuring graceful degradation and high availability.

XAI Framework for Smart Agriculture | *Python, XGBoost, Random Forest, RNN, CNN, ANN, IoT Analytics*

- Developed an Explainable AI system for smart agriculture covering crop recommendation, yield forecasting, and rainfall prediction using ensemble and deep learning models including Decision Tree, Random Forest, XGBoost, AdaBoost, SVC, RNN, CNN, and ANN.
- Built a real-time IoT analytics layer for continuous field monitoring, enabling data-driven insights that improved crop productivity and reduced resource waste.
- Applied preprocessing pipelines including feature selection, normalization, and cross-validation for robust model performance, evaluated via accuracy and F1-score metrics.
- Integrated XAI techniques to provide transparent, interpretable model explanations accessible to non-technical end users.

Liquid Neural Networks for Bipedal Robot Navigation | *Python, ROS2, LTCNs, LSTM, Transformers, Deep RL*

- Developed a navigation and obstacle avoidance system for bipedal robots in dynamic 2D environments using Liquid Time-Constant Networks (LTCNs), LSTMs, and Transformers.
- Implemented deep reinforcement learning simulations using ROS2 on Ubuntu 22.04 and Bipedal Walker, with LTCNs demonstrating superior adaptability over competing architectures.
- Evaluated model performance using MSE and cumulative reward metrics, with results validating LTCNs as the most effective architecture for real-world robotic autonomy.